Atmosic AT Command for ATM2/ATM3 Wireless SoC Family User Guide

Revision History

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July 1, 2021	0.50	Initial version created.
September 29, 2021	0.51	Added section 2.23 to section 2.32
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1 Overview

This document describes how to control the Atmosic Evaluation Kit (**ATM2/ATM3** 5x5 mm package or 6x6 mm package) using the AT commands. The default setting of the Atmosic AT command firmware is as below:

UART0:
Flow Control: CTS/RTS is mandatory Baud Rate: 460800

Legacy advertising Parameter:
Interval minimal: 20 ms (0x00 20)
Interval maximal: 40 ms (0x00 40)
Duration: infinite (0x00)

Advertising data:
Complete Local Name: Atmosic ATCmd
Shortened Local Name: AT
(08 FF 24 0A 41 54 43 4D 44 0E 09 41 74 6D 6F 73 69 63 20 41 54 43 6D 64 03 08 41 54)

Scan response data:
Complete Local Name: Atmosic ATCmd
Shortened Local Name: AT
(08 FF 24 0A 41 54 43 4D 44 0E 09 41 74 6D 6F 73 69 63 20 41 54 43 6D 64 03 08 41 54)

Security property:
Service: No Security (0x00 00)
Characteristic: No Security, Write Request and Notification (0x18 00)

Figure 1-1 shows the hardware configuration based on the EVB (evaluation board) and interface board.

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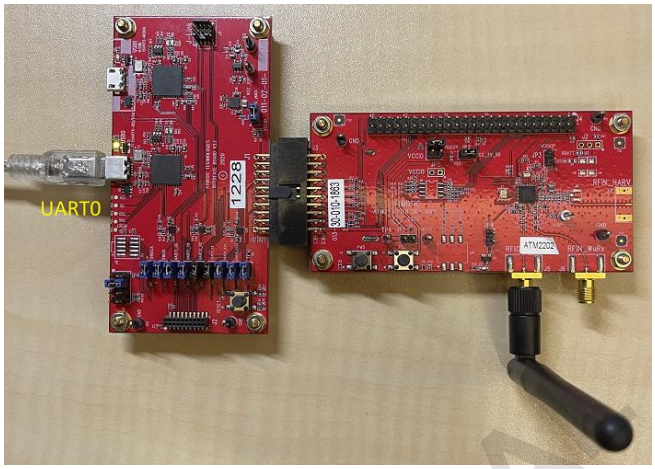


Figure 1-1 - Hardware Configuration with EVB (Left) and Interface Board (Right)

1.1 Quick Start

Enable Default GATT Server

AT+BLEGATTDFTSERVER=ON
AT+SYSRESET=00

Enable advertising

AT+BLEGAPINIT=0F
AT+BLEADVENABLE=00,ON

Use the nRF *Connect* app to scan and see the device in the list.

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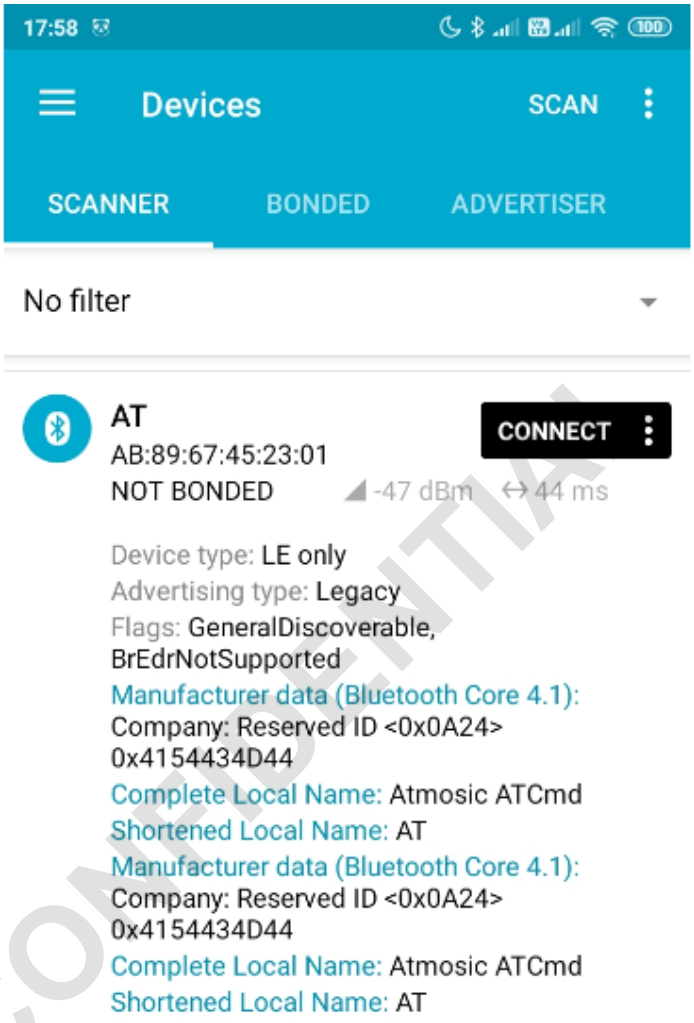
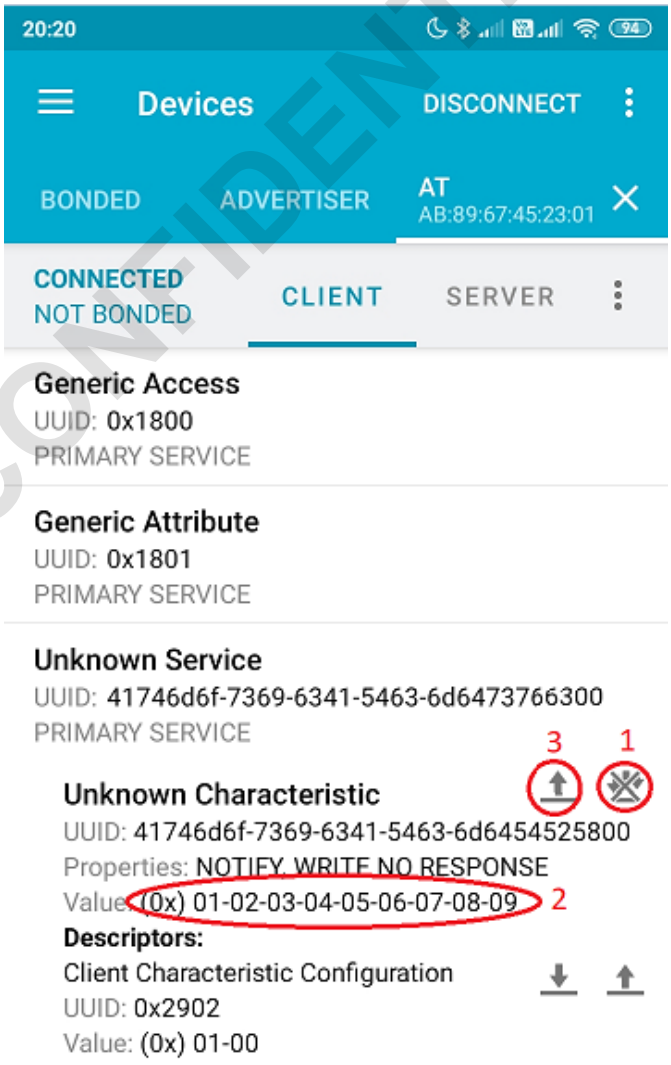


Figure 1.1-1 - nRF Connect App

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Connect to device via *nRF Connect* app

1. Click the "*CONNECT*" button in the *nRF Connect* App to make the Bluetooth LE connection between smartphone and device.
2. The device will send an event "*+BLEGAPCONN:00*" to the terminal that indicates the connection was established.
3. Please click "1" to enable CCCD for data transmission between nRF Connect App and Atmosic EVK.
4. Subsequently, apply the transmit data command to send data to the nRF *Connect* App as follows, "*AT+BLEGATTTOBLE=00,01,02,010203040506070809*" and the result as shown in Figure 1.1-2.
5. Press the button "3' to use the *nRF Connect* App sends data (0x1234) to the device then will show the result in the terminal in Figure 1.1-3, "*+BLEGATTFROMBLE:00,01,01,1234*".



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Figure 1.1-2 - Send Data to nRF Connect App

```
AT+BLEGATTDFTSERVER=ON
+BLEGATTDFTSERVER:4
OK

AT+SYSRESET=00

AT+BLEGAPINIT=0F
OK

AT+BLEADVENABLE=0,ON
OK

+BLEGAPCONN:0

AT+BLEGATTTOBLE=00,01,02,010203040506070809
OK
+BLEGATTTOBLE:0,02,6,1

+BLEGATTFROMBLE:0,1,01,1234
```

Figure 1.1-3 - Terminal Result

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2 AT Command List

2.1 AT+BLEGATTDFTSERVER

Turn ON or OFF the default GATT server in system booting up	
AT+BLEGATTDFTSERVER=<action>	The setting will be applied after system reset. <action>, ON or OFF <return>, AT_SYS_NEED_REBOOT (4)

Result:

```
AT+BLEGATTDFTSERVER=?
+BLEGATTDFTSERVER:<ON|OFF>
OK

AT+BLEGATTDFTSERVER?
+BLEGATTDFTSERVER:ON
OK

AT+BLEGATTDFTSERVER=OFF
+BLEGATTDFTSERVER:4
OK

AT+SYSRESET=00

AT+BLEGATTDFTSERVER?
+BLEGATTDFTSERVER:OFF
OK
```

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2.2 AT+BLEGATTDFTSERVICE

Set the UUID of the default service, and the default ATT server is available.	
AT+BLEGATTDFTSERVICE= <sec_prop>, <uuid>	The setting will be applied after system reset. <sec_prop> 0000: No security 0004: Unauthenticate 0008: Authenticate 000C: Security connection <uuid>, the service UUID

Result:

```
AT+BLEGATTDFTSERVICE=?
+BLEGATTDFTSERVICE:<sec_prop>,<uuid>
OK

AT+BLEGATTDFTSERVICE=0000,41746D6F736963000000000000000F1
OK

AT+BLEGATTDFTSERVICE?
+BLEGATTDFTSERVICE:0000,41746D6F736963000000000000000F1
OK
```

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2.3 AT+BLEGATTDFTCHAR1

Set the UUID of the default characteristic 1, and the default ATT server is available.	
AT+BLEGATTDFTCHAR1= <sec_prop>, <uuid>	The setting will be applied after system reset. <sec_prop>, see below table. <uuid>, the characteristic UUID

If the default ATT server is available,

```
*   15   14   13   12   11   10    9    8    7    6    5    4    3    2    1    0
* +----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+
* | X | WS |  I  | N  | WR | WC | RD | X  |    NP   |    IP   |   WP    |    RP   |
* +----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+
*
* Bit [0-1]  : Read Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [2-3]  : Write Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [4-5]  : Indication Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [6-7]  : Notification Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
*
* Bit [8]    : Reserved
* Bit [9]    : Read Command accepted
* Bit [10]   : Write Command accepted
* Bit [11]   : Write Request accepted
* Bit [12]   : Notification
* Bit [13]   : Indication
* Bit [14]   : Write Signed accepted
```

Result:

```
AT+BLEGATTDFTCHAR1=?
+BLEGATTDFTCHAR1:<sec_prop>,<uuid>
OK

AT+BLEGATTDFTCHAR1=1800,41746D6F7369630000000000000000F2
OK
```

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```
AT+BLEGATTDFTCHAR1?
+BLEGATTDFTCHAR1:1800,41746D6F7369630000000000000000F2
OK
```

2.4 AT+BLEGATTDFTCHAR2

Set the UUID of the default characteristic 2, and the default ATT server is available.	
AT+BLEGATTDFTCHAR2= <sec_prop>, <uuid>	The setting will be applied after system reset. <sec_prop>, see below table. <uuid>, the characteristic UUID

If the default ATT server is available,

```
*   15   14   13   12   11   10    9    8    7    6    5    4    3    2    1    0
* +----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+
* | X | WS |  I  | N  | WR | WC | RD | X  |    NP  |    IP   |   WP    |    RP   |
* +----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+
*
* Bit [0-1]  : Read Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [2-3]  : Write Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [4-5]  : Indication Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [6-7]  : Notification Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
*
* Bit [8]    : Reserved
* Bit [9]    : Read Command accepted
* Bit [10]   : Write Command accepted
* Bit [11]   : Write Request accepted
* Bit [12]   : Notification
* Bit [13]   : Indication
* Bit [14]   : Write Signed accepted
```

Result:

```
AT+BLEGATTDFTCHAR2=?
+BLEGATTDFTCHAR2:<sec_prop>,<uuid>
OK
```

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```
AT+BLEGATTDFTCHAR2=1800,41746D6F7369630000000000000000F3
OK

AT+BLEGATTDFTCHAR2?
+BLEGATTDFTCHAR2:1800,41746D6F7369630000000000000000F3
OK
```

2.5 AT+BLEADVLEGACYPARM

Set advertising parameters.	
AT+BLEADVLEGACYPARM= <adv_idx>, <interval_min>, <interval_max>, <duration>	The setting will be applied after enabling advertising. <adv_idx>, advertising index from 0 to max supported number. <interval_min>, the minimal advertising interval value from 32 to 16384. (unit: 0.625 ms) <interval_max>, the maximal advertising interval value from 32 to 16384. (unit: 0.625 ms) <duration>, how long the advertising is available, and the value from 0 to 65535. (unit: 10 ms). P.S. Set this value to 0 to keep advertising..

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Result:

```
AT+BLEADVLEGACYPARM=?
+BLEADVLEGACYPARM:<adv_idx>,<interval_min>,<interval_max>,<duration>
OK

AT+BLEADVLEGACYPARM?
+BLEADVLEGACYPARM:0,64,64,0
+BLEADVLEGACYPARM:1,64,64,0
OK

AT+BLEADVLEGACYPARM=0,32,64,10
OK
```

```
AT+BLEADVLEGACYPARM=1,64,128,100
OK

AT+BLEADVLEGACYPARM?
+BLEADVLEGACYPARM:0,32,64,10
+BLEADVLEGACYPARM:1,64,128,100
OK
```

2.6 AT+BLEADVDATA

Set advertising data.	
AT+BLEADVDATA= <adv_idx>, <advertising_data>	<adv_idx>, advertising index from 0 to max supported number. <advertising_data>, advertising data, the maximum is 31 bytes

Result:

```
AT+BLEADVDATA=?
+BLEADVDATA:<act_idx>,<advertising data>
OK

AT+BLEADVDATA?
+BLEADVDATA:0,08FF240A4154434D440E0941746D6F736963204154436D6403084154
+BLEADVDATA:1,08FF240A4154434D440E0941746D6F736963204154436D6403086174
OK

AT+BLEADVDATA=0,050841424344
OK

AT+BLEADVDATA=1,0308414102010612FF4C000631010065F3102C0B06000100010204511BA7
04
OK

AT+BLEADVDATA?
+BLEADVDATA:0,050841424344
+BLEADVDATA:1,0308414112FF4C000631010065F3102C0B06000100010204511BA704
OK
```

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2.7 AT+BLESCANRSPDATA

Set scan response data	
AT+BLESCANRSPDATA= <adv_idx>, <scan_response>	<adv_idx>, advertising index from 0 to max supported number. <advertising_data>, scan response data, the maximum is 31 bytes

Result:

```
AT+BLESCANRSPDATA=?
+BLESCANRSPDATA:<act_idx>,<scan response>
OK

AT+BLESCANRSPDATA?
+BLESCANRSPDATA:0,08FF240A4154434D440E0941746D6F736963204154436D6403084154
+BLESCANRSPDATA:1,08FF240A4154434D440E0941746D6F736963204154436D6403086174
OK

AT+BLESCANRSPDATA=0,03084154
OK
AT+BLESCANRSPDATA=1,0E0941746D5F736963204154436D64
OK

AT+BLESCANRSPDATA?
+BLESCANRSPDATA:0,03084154
+BLESCANRSPDATA:1,0E0941746D5F736963204154436D64
OK
```

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2.8 AT+BLEADVENABLE

Enable advertising feature	
AT+BLEADVENABLE= <adv_idx>, <action>	The GAP must be initialized (2.18 AT+BLEGAPINIT) before applying this setting. <adv_idx>, advertising index from 0 to max supported number. <action>, ON or OFF Response Code: ATM_ADV_IDLE (0) ATM_ADV_CREATING (1) ATM_ADV_CREATED (2) ATM_ADV_ADVDATA_SETTING (3) ATM_ADV_ADVDATA_DONE (4) ATM_ADV_SCANDATA_SETTING (5) ATM_ADV_SCANDATA_DONE (6) ATM_ADV_OFF (7) ATM_ADV_STARTING (8) ATM_ADV_ON (9) ATM_ADV_STOPPING (10) ATM_ADV_DELETING (11) ATM_ADV_DELETED (12)

Result:

```
AT+BLEADVENABLE=?
+BLEADVENABLE:<act_idx>,<ON|OFF>
OK

AT+BLEADVENABLE=0,ON
+BLEADVENABLE:72
ERR

AT+BLEGAPINIT=0F
OK

AT+BLEADVENABLE?
+BLEADVENABLE:0,0
+BLEADVENABLE:1,0
OK
```

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```
AT+BLEADVENABLE=0,ON
OK
AT+BLEADVENABLE=1,ON
OK

AT+BLEADVENABLE?
+BLEADVENABLE:0,9
+BLEADVENABLE:1,9
OK
```

2.9 AT+BLEGAPDISCONNECT

Disconnect the BLE connection by connection index	
AT+BLEGAPDISCONNECT=<conn_idx>	<conn_idx>, disconnect the current connection. <conn_idx> returns from the *BLEGATTFROMBLE* event after the connection is established.

Result:

```
AT+BLEGAPDISCONNECT=?
+BLEGAPDISCONNECT:<conn_idx>
OK

AT+BLEGAPINIT=0F
OK

AT+BLEADVENABLE=0,ON
OK

+BLEGAPCONN:0

AT+BLEGAPDISCONNECT=0
OK

+BLEGAPDISCONN:0,22
```

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2.10 AT+BLEGATTTOBLE

Transmit data to the remote through BLE after connection is established.	
AT+BLEGATTTOBLE= <conn_idx>, <att_idx>, <action>, <data>	<conn_idx>, the <conn_idx> returns from the BLEGAPCONN event after the connection is established. <att_idx>, it uses the static <att_idx> if the default ATT server is available. 01: Characteristic 1 03: Characteristic 2 <action> 00: Read 01: Write 02: Notification 03: Indication 04: Read Confirm 05: ATT info 06: ATT info Confirm 07: Notification (Empty) 08: Indication (Empty)

Result:

The default ATT server is OFF in this example.
Note: Please enable CCCD on the remote side before applying the notification.

```
AT+BLEGATTDFTSERVER=ON
+BLEGATTDFTSERVER:4
OK

AT+SYSRESET=00

AT+BLEGAPINIT=0F
OK

AT+BLEADVENABLE=0,ON
OK

+BLEGAPCONN:0

AT+BLEGATTTOBLE=00,01,02,010203040506070809
OK
```

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```
+BLEGATTTOBLE:0,02,6,1
```

2.11 AT+LISTCMDS

List all AT commands.	
AT+LISTCMDS?	Returns a list of all AT commands, see Result below.

Result:

```
AT+LISTCMDS?

+BLEADVDATA:<act_idx>,<advertising data>
+BLEADVENABLE:<act_idx>,<ON|OFF>
+BLEADVLEGACYPARM:<adv_idx>,<interval_min>,<interval_max>,<duration>
+BLEGAPDEVNAME:<device name>
+BLEGAPDISCONNECT:<conn_idx>
+BLEGAPINIT:<role>
+BLEGATTADDCHAR:<sec_prop>,<uuid>,<max_size>
+BLEGATTADDDESC:<uuid>
+BLEGATTADDDESCUSERDFD:<sec_prop>,<uuid>,<max_size>
+BLEGATTADDSERVICE:<sec_prop>,<uuid>
+BLEGATTDFTCHAR1:<sec_prop>,<uuid>
+BLEGATTDFTCHAR2:<sec_prop>,<uuid>
+BLEGATTDFTSERVER:<ON|OFF>
+BLEGATTDFTSERVICE:<sec_prop>,<uuid>
+BLEGATTMTUEXCHGREQ:<conn_idx>
+BLEGATTMTUSET:<mtu>
+BLEGATTSVCACTIVE:<reason>
+BLEGATTTOBLE:<conn_idx>,<att_idx>,<action>,<data>
+BLESCANRSPDATA:<act_idx>,<scan response>
+CUSTDEMO:<action>
+DEBUG:<1 or 0>
+LISTCMDS:<List all AT commands>
+SYSBDADDR:<Address>
+SYSRESET:<reason>
OK
```

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2.12 AT+SYSBDADDR

Modify the BD address of the device.	
AT+SYSBDADDR=<address>	<address>, BD address of the device. <return>, AT_GAP_NEED_SET_BEFORE_GAP_INIT (3)

Result:

```
AT+SYSBDADDR=?
+SYSBDADDR:<Address>
OK

AT+SYSBDADDR=00904B010203
OK

AT+SYSBDADDR?
+SYSBDADDR:00904B010203
OK

AT+SYSBDADDR=40904B010203
OK

AT+SYSBDADDR?
+SYSBDADDR:40904B010203
OK

AT+BLEGAPINIT=0F
OK

AT+SYSBDADDR=FF904B010203
+SYSBDADDR:3
OK
```

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2.13 AT+BLEGATTADDSERVICE

Add service (dynamic)	
AT+BLEGATTADDSERVICE= <sec_prop>, <uuid>	The service will be activated by GAP initial command (2.18 AT+BLEGAPINIT). Apply the active service (2.16 AT+BLEGATTSVCACTIVE) command if the service is added after GAP initializes. <sec_prop> 0000: No security 0004: Unauthenticate 0008: Authenticate 000C: Security connection <uuid>, the service UUID <return> att_idx

Result:

```
AT+BLEGATTADDSERVICE=?
+BLEGATTADDSERVICE:<sec_prop>,<uuid>
OK

AT+BLEGATTADDSERVICE=0000,41746D6F7369630000000000000000001
+BLEGATTADDSERVICE:0
OK
```

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2.14 AT+BLEGATTADDCHAR

Add characteristic (dynamic)	
AT+BLEGATTADDCHAR= <sec_prop>, <uuid>, <max_size>	<sec_prop>, see below table. <uuid>, the characteristic UUID <max_size> 1 ~ 1024 <return> att_idx

```
*   15   14   13   12   11   10    9    8    7    6    5    4    3    2    1    0
* +----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+
* | X | WS |  I  | N  | WR | WC | RD | X  |    NP   |    IP   |   WP    |    RP   |
* +----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+
*
* Bit [0-1]  : Read Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [2-3]  : Write Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [4-5]  : Indication Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [6-7]  : Notification Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
*
* Bit [8]    : Reserved
* Bit [9]    : Read Command accepted
* Bit [10]   : Write Command accepted
* Bit [11]   : Write Request accepted
* Bit [12]   : Notification
* Bit [13]   : Indication
* Bit [14]   : Write Signed accepted
```

Result:

```
AT+BLEGATTADDCHAR=?
+BLEGATTADDCHAR:<sec_prop>,<uuid>,<max_size>
OK

AT+BLEGATTADDCHAR=1800,41746D6F736963000000000000F0002,512
+BLEGATTADDCHAR:2
OK
```

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2.15 AT+BLEGATTADDDESC

Add descriptor (dynamic)	
AT+BLEGATTADDDESC= <uuid>	<uuid> 2900: Characteristic extended properties descriptor 2902: Client characteristic configuration descriptor 2903: Server characteristic configuration descriptor 2904: Characteristic presentation format descriptor <return> att_idx

Result:

```
AT+BLEGATTADDDESC=?
+BLEGATTADDDESC:<uuid>
OK

AT+BLEGATTADDSERVICE=0000,41746D6F736963000000000000000001
+BLEGATTADDSERVICE:0
OK

AT+BLEGATTADDCHAR=1800,41746D6F736963000000000000F0002,512
+BLEGATTADDCHAR:2
OK

AT+BLEGATTADDDESC=2902
+BLEGATTADDDESC:3
OK
```

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2.16 AT+BLEGATTSVCACTIVE

Active service (dynamic)	
AT+BLEGATTSVCACTIVE=<reason>	<reason> 00: Apply

Result:

```
AT+BLEGATTSVCACTIVE=?
+BLEGATTSVCACTIVE:<reason>
OK

AT+BLEGAPINIT=0F
OK

AT+BLEGATTADDSERVICE=0000,41746D6F736963000000000000000F1
+BLEGATTADDSERVICE:0
OK

AT+BLEGATTADDCHAR=1800,41746D6F736963000000000000000F2,512
+BLEGATTADDCHAR:2
OK

AT+BLEGATTADDDESC=2902
+BLEGATTADDDESC:3
OK

AT+BLEGATTSVCACTIVE=00
OK
```

CONFIDENTIAL

2.17 AT+SYSRESET

Reset system (software)	
AT+SYSRESET= <reason>	<reason> 00: No error

Result:

```
AT+SYSRESET=?
+SYSRESET:<reason>
OK

AT+SYSRESET=00
```

CONFIDENTIAL

2.18 AT+BLEGAPINIT

BLE GAP initialization	
AT+BLEGAPINIT= <role>	This command could only apply once, and needs to be reset if it changes to another role. <role> 01: Observer 02: Broadcaster 05: Central 0A: Peripheral 0F: All role

Result:

```
AT+BLEGAPINIT=?
+BLEGAPINIT:<role>
OK

AT+BLEGAPINIT?
+BLEGAPINIT:0
OK

AT+BLEGAPINIT=0F
OK

AT+BLEGAPINIT?
+BLEGAPINIT:0xf
OK
```

CONFIDENTIAL

2.19 BLEGATTADDDESCUSERDFD

System power mode control	
AT+BLEGATTADDDESCUSERDFD= <sec_prop>, <uuid>, <max_size>	<sec_prop>, see below table. <uuid>, the characteristic UUID <max_size> 1 ~ 1024 <return> att_idx

```
*   15   14   13   12   11   10    9    8    7    6    5    4    3    2    1    0
* +----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+
* | X | WS |  I  | N  | WR | WC | RD | X  |    NP   |    IP   |   WP    |    RP   |
* +----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+----+
*
* Bit [0-1]  : Read Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [2-3]  : Write Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [4-5]  : Indication Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
* Bit [6-7]  : Notification Permission
               (0 = NO_AUTH, 1 = UNAUTH, 2 = AUTH, 3 = SEC_CON)
*
* Bit [8]    : Reserved
* Bit [9]    : Read Command accepted
* Bit [10]   : Write Command accepted
* Bit [11]   : Write Request accepted
* Bit [12]   : Notification
* Bit [13]   : Indication
* Bit [14]   : Write Signed accepted
```

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Result:

```
AT+BLEGATTADDDESCUSERDFD=?
+BLEGATTADDDESCUSERDFD:<sec_prop>,<uuid>,<max_size>
OK

AT+BLEGATTADDSERVICE=0000,41746D6F736963000000000000000F1
+BLEGATTADDSERVICE:0
OK

AT+BLEGATTADDCHAR=1800,41746D6F736963000000000000000F2,512
+BLEGATTADDCHAR:2
OK

AT+BLEGATTADDDESCUSERDFD=0A00,41746D6F7369630000000000000008,8
+BLEGATTADDDESCUSERDFD:3
OK

AT+BLEGAPINIT=0F
OK

AT+BLEADVENABLE=00,ON
OK

# use nRF Connect app to make connection
+BLEGAPCONN:0

# use nRF Connect app to read descriptor
+BLEGATTFROMBLE:0,3,00,

AT+BLEGATTTOBLE=0,3,04,0102030405060708

OK
```

CONFIDENTIAL

2.20 BLEGATTMTUSET

Set ATT MTU size	
AT+BLEGATTMTUSET= <mtu>	<conn_idx>, connection index <mtu>, ATT MTU size form 23 to 515 <return>, AT_GAP_NEED_SET_BEFORE_GAP_INIT (3)

Result:

```
AT+BLEGATTMTUSET=?
+BLEGATTMTUSET:<mtu>
OK

AT+BLEGATTMTUSET=111
OK

AT+BLEGAPINIT=0F
OK

AT+BLEGATTMTUSET=222
+BLEGATTMTUSET:3
ERR
```

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2.21 BLEGATTMTUEXCHGREQ

Apply the MTU exchange request (only available once per connection)	
AT+BLEGATTMTUEXCHGREQ=<conn_idx>	<conn_idx>, connection index

Result:

```
AT+BLEGATTMTUEXCHGREQ=?
+BLEGATTMTUEXCHGREQ:<conn_idx>
OK

AT+SYSBDADDR=40904B010203
OK

AT+BLEGATTMTUSET=111
OK

AT+BLEGAPINIT=0F
OK

AT+BLEADVENABLE=0,ON
OK

+BLEGAPCONN:0

AT+BLEGATTMTUEXCHGREQ=0
OK
+BLEGATTMTUEXCHANGE:0,111
+BLEGATTMTUEXCHGREQ:0
```

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2.22 BLEGAPDEVNAME

Set ATT MTU size	
AT+BLEGAPDEVNAME=<device name>	<device name>, GAP device name and the maximum length is 18 characters. <return>, AT_GAP_NEED_SET_BEFORE_GAP_INIT (3)

Result:

```
AT+BLEGAPDEVNAME=?
+BLEGAPDEVNAME:<device name>
OK

AT+BLEGAPDEVNAME?
+BLEGAPDEVNAME:Atmosic ATCmd
OK

AT+BLEGAPDEVNAME=AT Command
OK

AT+BLEGAPDEVNAME?
+BLEGAPDEVNAME:AT Command
OK

AT+BLEGAPINIT=0F
OK

AT+BLEGAPDEVNAME=Test
+BLEGAPDEVNAME:3
ERR
```

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2.23 SYSUART0BR

Set UART0 baud rate	
AT+SYSUART0BR= <baud rate>, <defer apply>	<baud rate>, new baud rate setting <defer apply>, the setting will be applied after the delay of 1~255 centi seconds.

Result:

```
AT+SYSUART0BR=?
+SYSUART0BR:<baud rate>,<defer apply>
OK

AT+SYSUART0BR?
+SYSUART0BR:460800
OK

AT+SYSUART0BR=115200,10
OK
```

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2.24 SYSUART0FC

Set UART0 flow control	
AT+SYSUART0FC= <action>	<action> 0: Flow Control OFF 1: Flow Control ON

Result:

```
AT+SYSUART0FC=?
+SYSUART0FC:<action>
OK

AT+SYSUART0FC?
+SYSUART0FC:1
OK

AT+SYSUART0FC=0
OK

AT+SYSUART0FC?
+SYSUART0FC:0
OK
```

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2.25 SYSFUNCPIN

Function pin control	
AT+SYSFUNCPIN= <io>, <direction>, <signal>	<pio>, pin number <direction> 00: Input 01: Output <signal> 00: Low level 01: High level 02: Rising edge 03: Falling edge 04: Pull up (Input Only)

Result:

```
+SYSFUNCPIN:<io>,<direction>,<signal>
OK

AT+SYSFUNCPIN=9,0,2
OK

# Notification after trigger the P9
+SYSFUNCPIN:9,4
```

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2.26 SYSPM

Power mode setting	
AT+SYSPM= <mode>, <io>, <wakeup level>	<mode> 01: Hibernation mode <io>, pio number <wakeup level> 00: Low level 01: High level

Result:

```
AT+SYSPM=?
+SYSPM:<mode>,<io>,<wakeup level>
OK

AT+SYSPM=1,9,0
OK

# System into hibernation mode till trigger the P9
```

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2.27 BLEADVTXPWR

Set advertising Tx power	
AT+BLEADVTXPWR= <dBm>	<dBm>, support -20, -10, -6, -4, -2, 0, 2 and 4 dBm

Result:

```
AT+BLEADVTXPWR=?
+BLEADVTXPWR:<dBm>
OK

AT+BLEADVTXPWR=0
OK
```

2.28 BLECONNTXPWR

Set connection Tx power (*Only available after BLE connection established*)	
AT+BLECONNTXPWR= <conn_idx>, <dBm>	<conn_idx>, connection index <dBm>, support -20, -10, -6, -4, -2, 0, 2 and 4 dBm

Result:

```
AT+BLECONNTXPWR=?
+BLECONNTXPWR:<conn_idx>,<dBm>
OK

# After BLE connection established
AT+BLECONNTXPWR=0,0
OK
```

2.29 BLEGAPGETCONNSTAT

Get connection state	
AT+BLEGAPGETCONNSTAT= <conn_idx>	<conn_idx>, connection index <state> LINK_CLOSED (0), LINK_CONNECTING (1), LINK_CONNECTED (2), LINK_CLOSING (3),

Result:

```
AT+BLEGAPGETCONNSTAT=?
+BLEGAPGETCONNSTAT:<conn_idx>
OK

# After BLE connection established
AT+BLEGAPGETCONNSTAT=0
+BLEGAPGETCONNSTAT:2
OK
```

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2.30 BLEGAPGETRSSI

Get connection RSSI (*Only available after BLE connection established*)	
AT+BLEGAPGETRSSI= <conn_idx>	<conn_idx>, connection index

Result:

```
AT+BLEGAPGETRSSI=?
+BLEGAPGETRSSI:<conn_idx>
OK

AT+BLEGAPGETRSSI=0
OK
+BLEGAPLINFO:0 RSSI (-71)
```

2.31 SYSUART0RAW

Set AT command to raw mode	
AT+SYSUART0RAW= <conn_idx>, <att_idx>, <direction>, <length>	<conn_idx>, connection index <att_idx>, attribute index <direction>, to/from Atmosic chip 0: Host to Atmosic chip 1: Atmosic to host <length>, data length in bytes

Result:

```
AT+SYSUART0RAW=?
+SYSUART0RAW:<conn_idx>,<att_idx>,<direction>,<length>
OK

AT+SYSUART0RAW=0,0,0,12
OK

# System will switch AT command after Host sending 12 bytes to Atmosic
```

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2.32 SYSDFU

DFU mode (*System will go into raw mode after applying this command*)	
AT+SYSDFU= <type>, <address>, <length>, <crc>	<type>, content type 0: Firmware (.bin) 1: NVDS (.bin) <address>, hex format (4 bytes) Firmware: inactive partition offset NVDS: NVDS offset <length>, the content data length, hex format (4 bytes) <crc> 0: crc disable 1: crc enable <return> /// No error AT_CMD_DFU_ERR_NO_ERROR (0), /// Sequence number error AT_CMD_DFU_ERR_SN (1), /// Crc error AT_CMD_DFU_ERR_CRC (2), /// No resource AT_CMD_DFU_ERR_NO_RESOOURCE (3), /// Wrong header AT_CMD_DFU_ERR_WRONG_HDR (4), /// Wrong address AT_CMD_DFU_ERR_WRONG_ADDR (5), /// BLE is working AT_CMD_DFU_ERR_BLE_WORKING (6),
AT+SYSDFU?	Query flash information: 0: Capacity 1: Partition size 2: Active partition offset 3: Active partition status 4: Inactive partition offset 5: Inactive partition status 6: NVDS partition size 7: NVDS partition offset

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Result:

```
AT+SYSDFU=?
+SYSDFU:<type>,<address>,<length>,<crc>
OK

AT+SYSDFU?
# capacity
+SYSDFU:0,80000
# partition size
+SYSDFU:1,40000
# active partition offset
+SYSDFU:2,0
# active partition status
+SYSDFU:3,1
# inactive partition offset
+SYSDFU:4,40000
# inactive partition status
+SYSDFU:5,1
# NVDS partition size
+SYSDFU:6,8000
# NVDS partition offset
+SYSDFU:7,78000
OK

# Into DFU mode
# Content: firmware, address: 0x40000, length: 0x142E0, CRC enable
AT+SYSDFU=0,00040000,000142E0,1
# Only raw data available after Host sending 0x142E0 bytes to Atmosic,
# or there is an error occur
```

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2.32.1 CRC

The CRC checks configuration, Width=8, Poly=0x97, XorIn=0x00, ReflectIn=False, XorOut=0x00, ReflectOut=False, Algorithm=table-driven

please refer to *platform/atm2/ATM22xx-x0x/lib/crc/crc8.c*

```
static const crc8_t crc_table[256] = {
    0x00, 0x97, 0xb9, 0x2e, 0xe5, 0x72, 0x5c, 0xcb,
    0x5d, 0xca, 0xe4, 0x73, 0xb8, 0x2f, 0x01, 0x96,
    0xba, 0x2d, 0x03, 0x94, 0x5f, 0xc8, 0xe6, 0x71,
    0xe7, 0x70, 0x5e, 0xc9, 0x02, 0x95, 0xbb, 0x2c,
    0xe3, 0x74, 0x5a, 0xcd, 0x06, 0x91, 0xbf, 0x28,
    0xbe, 0x29, 0x07, 0x90, 0x5b, 0xcc, 0xe2, 0x75,
    0x59, 0xce, 0xe0, 0x77, 0xbc, 0x2b, 0x05, 0x92,
    0x04, 0x93, 0xbd, 0x2a, 0xe1, 0x76, 0x58, 0xcf,
    0x51, 0xc6, 0xe8, 0x7f, 0xb4, 0x23, 0x0d, 0x9a,
    0x0c, 0x9b, 0xb5, 0x22, 0xe9, 0x7e, 0x50, 0xc7,
    0xeb, 0x7c, 0x52, 0xc5, 0x0e, 0x99, 0xb7, 0x20,
    0xb6, 0x21, 0x0f, 0x98, 0x53, 0xc4, 0xea, 0x7d,
    0xb2, 0x25, 0x0b, 0x9c, 0x57, 0xc0, 0xee, 0x79,
    0xef, 0x78, 0x56, 0xc1, 0x0a, 0x9d, 0xb3, 0x24,
    0x08, 0x9f, 0xb1, 0x26, 0xed, 0x7a, 0x54, 0xc3,
    0x55, 0xc2, 0xec, 0x7b, 0xb0, 0x27, 0x09, 0x9e,
    0xa2, 0x35, 0x1b, 0x8c, 0x47, 0xd0, 0xfe, 0x69,
    0xff, 0x68, 0x46, 0xd1, 0x1a, 0x8d, 0xa3, 0x34,
    0x18, 0x8f, 0xa1, 0x36, 0xfd, 0x6a, 0x44, 0xd3,
    0x45, 0xd2, 0xfc, 0x6b, 0xa0, 0x37, 0x19, 0x8e,
    0x41, 0xd6, 0xf8, 0x6f, 0xa4, 0x33, 0x1d, 0x8a,
    0x1c, 0x8b, 0xa5, 0x32, 0xf9, 0x6e, 0x40, 0xd7,
    0xfb, 0x6c, 0x42, 0xd5, 0x1e, 0x89, 0xa7, 0x30,
    0xa6, 0x31, 0x1f, 0x88, 0x43, 0xd4, 0xfa, 0x6d,
    0xf3, 0x64, 0x4a, 0xdd, 0x16, 0x81, 0xaf, 0x38,
    0xae, 0x39, 0x17, 0x80, 0x4b, 0xdc, 0xf2, 0x65,
    0x49, 0xde, 0xf0, 0x67, 0xac, 0x3b, 0x15, 0x82,
    0x14, 0x83, 0xad, 0x3a, 0xf1, 0x66, 0x48, 0xdf,
    0x10, 0x87, 0xa9, 0x3e, 0xf5, 0x62, 0x4c, 0xdb,
    0x4d, 0xda, 0xf4, 0x63, 0xa8, 0x3f, 0x11, 0x86,
    0xaa, 0x3d, 0x13, 0x84, 0x4f, 0xd8, 0xf6, 0x61,
    0xf7, 0x60, 0x4e, 0xd9, 0x12, 0x85, 0xab, 0x3c
};
```

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2.32.2 Packet Format

The maximum packet length is 256 bytes and the format is as below.

```
The packet format (Host to Atmosic)
-----------------------------------------------------
|HDR(1)|SN(1)|LEN(1)|          DATA          |CRC(1)|
-----------------------------------------------------

/// The header of rdp packet
typedef struct at_cmd_rdp_hdr_s {
    union {
      uint8_t flag;
      struct {
          uint8_t type : 5;
          uint8_t op : 3;
      };
    };
    /// Sequence number
    uint8_t sn;
    /// Data length (exclude rdp header)
    uint8_t len;
} at_cmd_rdp_hdr_t;
```

HDR (Packet Header, 1 byte): The header consists of type and op code.

- type, the content information which 0 is firmware and 1 is NVDS.
- op
    - 0x0, the first packet and have more data
    - 0x1, the last packet
    - 0x2, the Ack. packet
    - 0x4, the continue packet

SN (Sequence Number, 1 byte): Start from 0 to 255
Length (N bytes): The maximum data length is *253* bytes w/o crc, *252* bytes w/ crc
CRC (1 byte): The CRC which is calculated from HDR to DATA

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2.32.3 Ack Packet Format

The maximum ack packet length is 3 bytes and the packet format is as below.

```
The ack packet format
---------------------
|HDR(1)|SN(1)|STS(1)|
---------------------
```

HDR (Packet Header, 1 byte): The header is the same as Section 2.32.2, and the op is 0x2
SN (Sequence Number, 1 byte): The SN should be the same as packet input
STS (Status, 1 byte):

- AT_CMD_DFU_ERR_NO_ERROR (**0**) - No error
- AT_CMD_DFU_ERR_SN (**1**) - Sequence number error
- AT_CMD_DFU_ERR_CRC (**2**) - CRC error
- AT_CMD_DFU_ERR_NO_RESOOURCE (**3**) - No resource
- AT_CMD_DFU_ERR_WRONG_HDR (**4**) - Wrong header
- AT_CMD_DFU_ERR_WRONG_ADDR (**5**) - Wrong address
- AT_CMD_DFU_ERR_BLE_WORKING (**6**) - BLE is working

2.32.4 Test Vector (Length 53 bytes)

2.32.4.1 Without CRC

Raw data (53 bytes)

```
13 48 14 49 14 4A 15 4B 9B 1A 0E C0 14 38 07 E0
FE 55 53 52 FF FF FF FF FF FF FF FF FF FF FF FF
80 21 09 06 01 60 00 21 01 60 00 23 43 60 0C 49
C1 60 0C 4A 52
```

DFU first and last packet

```
21 00 35 13 48 14 49 14 4A 15 4B 9B 1A 0E C0 14
38 07 E0 FE 55 53 52 FF FF FF FF FF FF FF FF FF
FF FF FF 80 21 09 06 01 60 00 21 01 60 00 23 43
60 0C 49 C1 60 0C 4A 52
```

Type is NVDS and op is the last packet.

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2.32.4.2 With CRC

DFU first and last packet

```
21 00 36 13 48 14 49 14 4A 15 4B 9B 1A 0E C0 14
38 07 E0 FE 55 53 52 FF FF FF FF FF FF FF FF FF
FF FF FF 80 21 09 06 01 60 00 21 01 60 00 23 43
60 0C 49 C1 60 0C 4A 52 37
```

Type is NVDS, op is the last packet and CRC is 0x37.

2.32.5 Test Vector (Length 293 bytes)

2.32.5.1 Without CRC

Raw data (293 bytes)

```
13 48 14 49 14 4A 15 4B 9B 1A 0E C0 14 38 07 E0
FE 55 53 52 FF FF FF FF FF FF FF FF FF FF FF FF
80 21 09 06 01 60 00 21 01 60 00 23 43 60 0C 49
C1 60 0C 4A 52 1A 02 61 0B 49 01 60 03 60 00 B5
0A F0 6C FC 01 BC 86 46 00 20 00 21 07 4B 18 47
08 00 00 70 D0 32 01 10 00 40 01 20 10 50 01 20
10 50 01 20 A4 5D 01 20 01 00 00 80 29 2D 00 10
70 47 70 47 13 B5 0A 78 03 00 00 92 02 49 03 4A
03 4C 04 48 A0 47 13 BD 70 E0 00 10 0A E1 00 10
49 66 00 00 50 E2 00 10 73 B5 0B 00 05 00 0C 00
08 4E 09 4A 09 49 0A 48 B0 47 0A 4B 22 00 18 78
00 23 29 00 01 93 00 93 07 4C 05 33 A0 47 80 20
76 BD C0 46 49 66 00 00 99 E0 00 10 70 E0 00 10
28 E2 00 10 00 40 01 20 69 1A 00 10 F0 B5 0C 00
16 00 1D 00 85 B0 03 90 00 93 09 4A 0B 00 09 4F
09 49 0A 48 B8 47 0A 4B 22 00 18 78 03 99 01 95
00 96 01 23 07 4C A0 47 00 20 05 B0 F0 BD C0 46
89 E0 00 10 49 66 00 00 70 E0 00 10 F8 E3 00 10
00 40 01 20 69
```

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DFU first packet (1)

```
01 00 FD 13 48 14 49 14 4A 15 4B 9B 1A 0E C0 14
38 07 E0 FE 55 53 52 FF FF FF FF FF FF FF FF FF
FF FF FF 80 21 09 06 01 60 00 21 01 60 00 23 43
60 0C 49 C1 60 0C 4A 52 1A 02 61 0B 49 01 60 03
60 00 B5 0A F0 6C FC 01 BC 86 46 00 20 00 21 07
4B 18 47 08 00 00 70 D0 32 01 10 00 40 01 20 10
50 01 20 10 50 01 20 A4 5D 01 20 01 00 00 80 29
2D 00 10 70 47 70 47 13 B5 0A 78 03 00 00 92 02
49 03 4A 03 4C 04 48 A0 47 13 BD 70 E0 00 10 0A
E1 00 10 49 66 00 00 50 E2 00 10 73 B5 0B 00 05
00 0C 00 08 4E 09 4A 09 49 0A 48 B0 47 0A 4B 22
00 18 78 00 23 29 00 01 93 00 93 07 4C 05 33 A0
47 80 20 76 BD C0 46 49 66 00 00 99 E0 00 10 70
E0 00 10 28 E2 00 10 00 40 01 20 69 1A 00 10 F0
B5 0C 00 16 00 1D 00 85 B0 03 90 00 93 09 4A 0B
00 09 4F 09 49 0A 48 B8 47 0A 4B 22 00 18 78 03
```

Type is NVDS and op is the first packet.

DFU continue and last packet (2)

```
A1 01 28 99 01 95 00 96 01 23 07 4C A0 47 00 20
05 B0 F0 BD C0 46 89 E0 00 10 49 66 00 00 70 E0
00 10 F8 E3 00 10 00 40 01 20 69
```

Type is NVDS and op is the last packet with continue.

2.32.5.2 With CRC

DFU first packet (1)

```
01 00 FD 13 48 14 49 14 4A 15 4B 9B 1A 0E C0 14
38 07 E0 FE 55 53 52 FF FF FF FF FF FF FF FF FF
FF FF FF 80 21 09 06 01 60 00 21 01 60 00 23 43
60 0C 49 C1 60 0C 4A 52 1A 02 61 0B 49 01 60 03
60 00 B5 0A F0 6C FC 01 BC 86 46 00 20 00 21 07
4B 18 47 08 00 00 70 D0 32 01 10 00 40 01 20 10
50 01 20 10 50 01 20 A4 5D 01 20 01 00 00 80 29
2D 00 10 70 47 70 47 13 B5 0A 78 03 00 00 92 02
49 03 4A 03 4C 04 48 A0 47 13 BD 70 E0 00 10 0A
E1 00 10 49 66 00 00 50 E2 00 10 73 B5 0B 00 05
00 0C 00 08 4E 09 4A 09 49 0A 48 B0 47 0A 4B 22
```

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```
00 18 78 00 23 29 00 01 93 00 93 07 4C 05 33 A0
47 80 20 76 BD C0 46 49 66 00 00 99 E0 00 10 70
E0 00 10 28 E2 00 10 00 40 01 20 69 1A 00 10 F0
B5 0C 00 16 00 1D 00 85 B0 03 90 00 93 09 4A 0B
00 09 4F 09 49 0A 48 B8 47 0A 4B 22 00 18 78 06
```

Type is NVDS, op is the first packet and the CRC is 0x06.

DFU continue and last packet (2)

```
A1 01 2A 03 99 01 95 00 96 01 23 07 4C A0 47 00
20 05 B0 F0 BD C0 46 89 E0 00 10 49 66 00 00 70
E0 00 10 F8 E3 00 10 00 40 01 20 69 B2
```

Type is NVDS, op is the last packet with continue and the CRC is 0xB2.

2.32.6 Test Vector (Length 525 bytes)

2.32.6.1 Without CRC

Raw data (525 bytes)

```
13 48 14 49 14 4A 15 4B 9B 1A 0E C0 14 38 07 E0
FE 55 53 52 FF FF FF FF FF FF FF FF FF FF FF FF
80 21 09 06 01 60 00 21 01 60 00 23 43 60 0C 49
C1 60 0C 4A 52 1A 02 61 0B 49 01 60 03 60 00 B5
0A F0 6C FC 01 BC 86 46 00 20 00 21 07 4B 18 47
08 00 00 70 D0 32 01 10 00 40 01 20 10 50 01 20
10 50 01 20 A4 5D 01 20 01 00 00 80 29 2D 00 10
70 47 70 47 13 B5 0A 78 03 00 00 92 02 49 03 4A
03 4C 04 48 A0 47 13 BD 70 E0 00 10 0A E1 00 10
49 66 00 00 50 E2 00 10 73 B5 0B 00 05 00 0C 00
08 4E 09 4A 09 49 0A 48 B0 47 0A 4B 22 00 18 78
00 23 29 00 01 93 00 93 07 4C 05 33 A0 47 80 20
76 BD C0 46 49 66 00 00 99 E0 00 10 70 E0 00 10
28 E2 00 10 00 40 01 20 69 1A 00 10 F0 B5 0C 00
16 00 1D 00 85 B0 03 90 00 93 09 4A 0B 00 09 4F
09 49 0A 48 B8 47 0A 4B 22 00 18 78 03 99 01 95
00 96 01 23 07 4C A0 47 00 20 05 B0 F0 BD C0 46
89 E0 00 10 49 66 00 00 70 E0 00 10 F8 E3 00 10
00 40 01 20 69 1A 00 10 73 B5 0B 00 06 00 0D 00
08 4A 09 49 09 4C 0A 48 A0 47 00 24 09 4B 2A 00
```

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```
18 78 31 00 08 4D 01 94 00 94 23 00 A8 47 20 00
76 BD C0 46 7A E0 00 10 70 E0 00 10 49 66 00 00
28 E2 00 10 00 40 01 20 69 1A 00 10 10 B5 03 00
04 4A 05 49 05 4C 06 48 A0 47 06 49 06 4B 00 20
98 47 10 BD C8 E0 00 10 70 E0 00 10 49 66 00 00
20 E3 00 10 EC E3 00 10 F5 9D 00 10 73 B5 0D 00
04 00 00 91 03 00 06 4E 06 4A 07 49 07 48 B0 47
07 4B 2A 00 18 78 21 00 06 4B 98 47 73 BD C0 46
49 66 00 00 AC E0 00 10 70 E0 00 10 BB E3 00 10
00 40 01 20 5D 1B 00 10 70 B5 04 00 03 00 04 4A
04 49 05 4D 05 48 A8 47 20 00 05 4B 98 47 70 BD
18 E1 00 10 70 E0 00 10 49 66 00 00 88 E2 00 10
C9 60 00
```

DFU first packet (1)

```
01 00 FD 13 48 14 49 14 4A 15 4B 9B 1A 0E C0 14
38 07 E0 FE 55 53 52 FF FF FF FF FF FF FF FF FF
FF FF FF 80 21 09 06 01 60 00 21 01 60 00 23 43
60 0C 49 C1 60 0C 4A 52 1A 02 61 0B 49 01 60 03
60 00 B5 0A F0 6C FC 01 BC 86 46 00 20 00 21 07
4B 18 47 08 00 00 70 D0 32 01 10 00 40 01 20 10
50 01 20 10 50 01 20 A4 5D 01 20 01 00 00 80 29
2D 00 10 70 47 70 47 13 B5 0A 78 03 00 00 92 02
49 03 4A 03 4C 04 48 A0 47 13 BD 70 E0 00 10 0A
E1 00 10 49 66 00 00 50 E2 00 10 73 B5 0B 00 05
00 0C 00 08 4E 09 4A 09 49 0A 48 B0 47 0A 4B 22
00 18 78 00 23 29 00 01 93 00 93 07 4C 05 33 A0
47 80 20 76 BD C0 46 49 66 00 00 99 E0 00 10 70
E0 00 10 28 E2 00 10 00 40 01 20 69 1A 00 10 F0
B5 0C 00 16 00 1D 00 85 B0 03 90 00 93 09 4A 0B
00 09 4F 09 49 0A 48 B8 47 0A 4B 22 00 18 78 03
```

DFU continue packet (2)

```
81 01 FD 99 01 95 00 96 01 23 07 4C A0 47 00 20
05 B0 F0 BD C0 46 89 E0 00 10 49 66 00 00 70 E0
00 10 F8 E3 00 10 00 40 01 20 69 1A 00 10 73 B5
0B 00 06 00 0D 00 08 4A 09 49 09 4C 0A 48 A0 47
00 24 09 4B 2A 00 18 78 31 00 08 4D 01 94 00 94
23 00 A8 47 20 00 76 BD C0 46 7A E0 00 10 70 E0
```

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```
00 10 49 66 00 00 28 E2 00 10 00 40 01 20 69 1A
00 10 10 B5 03 00 04 4A 05 49 05 4C 06 48 A0 47
06 49 06 4B 00 20 98 47 10 BD C8 E0 00 10 70 E0
00 10 49 66 00 00 20 E3 00 10 EC E3 00 10 F5 9D
00 10 73 B5 0D 00 04 00 00 91 03 00 06 4E 06 4A
07 49 07 48 B0 47 07 4B 2A 00 18 78 21 00 06 4B
98 47 73 BD C0 46 49 66 00 00 AC E0 00 10 70 E0
00 10 BB E3 00 10 00 40 01 20 5D 1B 00 10 70 B5
04 00 03 00 04 4A 04 49 05 4D 05 48 A8 47 20 00
05 4B 98 47 70 BD 18 E1 00 10 70 E0 00 10 49 66
```

DFU continue and last packet (3)

```
A1 02 09 00 00 88 E2 00 10 C9 60 00
```

2.32.6.2 With CRC

DFU first packet (1)

```
01 00 FD 13 48 14 49 14 4A 15 4B 9B 1A 0E C0 14
38 07 E0 FE 55 53 52 FF FF FF FF FF FF FF FF FF
FF FF 80 21 09 06 01 60 00 21 01 60 00 23 43
60 0C 49 C1 60 0C 4A 52 1A 02 61 0B 49 01 60 03
60 00 B5 0A F0 6C FC 01 BC 86 46 00 20 00 21 07
4B 18 47 08 00 00 70 D0 32 01 10 00 40 01 20 10
50 01 20 10 50 01 20 A4 5D 01 20 01 00 00 80 29
2D 00 10 70 47 70 47 13 B5 0A 78 03 00 00 92 02
49 03 4A 03 4C 04 48 A0 47 13 BD 70 E0 00 10 0A
E1 00 10 49 66 00 00 50 E2 00 10 73 B5 0B 00 05
00 0C 00 08 4E 09 4A 09 49 0A 48 B0 47 0A 4B 22
00 18 78 00 23 29 00 01 93 00 93 07 4C 05 33 A0
47 80 20 76 BD C0 46 49 66 00 00 99 E0 00 10 70
E0 00 10 28 E2 00 10 00 40 01 20 69 1A 00 10 F0
B5 0C 00 16 00 1D 00 85 B0 03 90 00 93 09 4A 0B
00 09 4F 09 49 0A 48 B8 47 0A 4B 22 00 18 78 06
```

DFU continue packet (2)

```
81 01 FD 03 99 01 95 00 96 01 23 07 4C A0 47 00
20 05 B0 F0 BD C0 46 89 E0 00 10 49 66 00 00 70
E0 00 10 F8 E3 00 10 00 40 01 20 69 1A 00 10 73
B5 0B 00 06 00 0D 00 08 4A 09 49 09 4C 0A 48 A0
```

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```
47 00 24 09 4B 2A 00 18 78 31 00 08 4D 01 94 00
94 23 00 A8 47 20 00 76 BD C0 46 7A E0 00 10 70
E0 00 10 49 66 00 00 28 E2 00 10 00 40 01 20 69
1A 00 10 10 B5 03 00 04 4A 05 49 05 4C 06 48 A0
47 06 49 06 4B 00 20 98 47 10 BD C8 E0 00 10 70
E0 00 10 49 66 00 00 20 E3 00 10 EC E3 00 10 F5
9D 00 10 73 B5 0D 00 04 00 00 91 03 00 06 4E 06
4A 07 49 07 48 B0 47 07 4B 2A 00 18 78 21 00 06
4B 98 47 73 BD C0 46 49 66 00 00 AC E0 00 10 70
E0 00 10 BB E3 00 10 00 40 01 20 5D 1B 00 10 70
B5 04 00 03 00 04 4A 04 49 05 4D 05 48 A8 47 20
00 05 4B 98 47 70 BD 18 E1 00 10 70 E0 00 10 1C
```

DFU continue and last packet (3)

```
A1 02 0C 49 66 00 00 88 E2 00 10 C9 60 00 E7
```

2.33 AT+BLESCANENABLE

Enable BLE scan	
AT+BLESCANENABLE= <idx>,<rssi>,<action>	The GAP must be initialized (2.18 AT+BLEGAPINIT) before applying this setting. <idx>, scanning index from 0 to max supported number. <rssi>, the minimum rssi value required. <action>, ON or OFF

Result:

```
AT+BLESCANENABLE=
<idx>,<rssi>,<ON|OFF>
OK

AT+BLESCANENABLE=0,-70,ON
OK
```

2.34 AT+BLESCANFILADD

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Add a BLE scan filter that only adv packet with matched string will be reported	
AT+BLESCANFILADD= <idx>,<ad_type>,<data>	The GAP must be initialized (2.18 AT+BLEGAPINIT) before applying this setting. <idx>, scanning index from 0 to max supported number. <ad_type>, the AD type field in the adv packet <data>, the string to be compared

Result:

```
AT+BLESCANFILADD=
<idx>,<ad_type>,<data>
OK

# filter adv packets that device name equals to <Atmosic RCU>
AT+BLESCANFILADD=0,09,41746d6F73696320524355
OK
```

2.35 AT+BLESCANFILDEL

Delete a specific BLE scan filter set previously	
AT+BLESCANFILDEL= <idx>,<ad_type>	The GAP must be initialized (2.18 AT+BLEGAPINIT) before applying this setting. <idx>, scanning index from 0 to max supported number. <ad_type>, the AD type field in the adv packet

Result:

```
AT+BLESCANFILDEL=
<idx>,<ad_type>
OK

# remove the filter of device name
AT+BLESCANFILADD=0,09
OK
```

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2.36 AT+BLESCANFILDELALL

Delete the BLE scan filter of all ad_types	
AT+BLESCANFILDEL= <idx>	The GAP must be initialized (2.18 AT+BLEGAPINIT) before applying this setting. <idx>, scanning index from 0 to max supported number.

Result:

```
AT+BLESCANFILDELALL=
<idx>
OK

AT+BLESCANFILADD=0
OK
```

2.37 AT+BLEGATTDISCSVCS

Discover services by handle range	
AT+BLEGATTDISCSVCS= <act_idx>, <start_handle>, <end_handle>, <max number>	<act_idx>, activity index <start_handle>, the starting handle to discover <end_handle>, the ending handle to discover <max number> maximum number of services to be discovered

2.38 AT+BLEGATTDISCSVC

Discover a specific service by UUID	
AT+BLEGATTDISCSVC= <act_idx>,<uuid>	<act_idx>, activity index <uuid> UUID of the service

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2.39 AT+BLEGATTDISCCHARS

Discover all characteristics in a range	
AT+BLEGATTDISCCHARS= <act_idx>, <start_handle>, <end_handle>, <max number>	<act_idx>, activity index <start_handle>, the starting handle to discover <end_handle>, the ending handle to discover <max number> maximum number of characteristics to be discovered

2.40 AT+BLEGATTDISCCHAR

Discover a specific characteristic by UUID	
AT+BLEGATTDISCCHAR= <act_idx>, <uuid>	<act_idx>, activity index <uuid> UUID of the characteristic

2.41 AT+BLEGATTDISCDESCS

Discover all descriptors in a range	
AT+BLEGATTDISCDESCS= <act_idx>, <start_handle>, <end_handle>, <max number>	<act_idx>, activity index <start_handle>, the starting handle to discover <end_handle>, the ending handle to discover <max number> maximum number of descriptors to be discovered

2.42 AT+BLEGATTWR

Write value to a characteristic and expect a response	
AT+BLEGATTWR= <act_idx>, <handle>,	<act_idx>, activity index <handle>, handle of the characteristic <data_array>, the data to be written

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<data_array>	

2.43 AT+BLEGATTWRWO

Write value to a characteristic without a expected response	
AT+BLEGATTWRWO= <act_idx>, <handle>, <data_array>	<act_idx>, activity index <handle>, handle of the characteristic <data_array>, the data to be written

2.44 AT+BLESMPPAIRREQ

Start the pairing procedure	
AT+BLESMPPAIRREQ= <conidx>, <iocap>, <oob>, <authreq>, <enckey_size>, <ikey_dist>, <rkey_dist>	<conidx>, connection index <iocap>, IO capabilities <oob>, OOB information <authreq>, authentication requirements <enckey_size>, Encryption key size (7 to 16) <ikey_dist>, Initiator key distribution flags <rkey_dist> Responder key distribution flags

2.45 AT+BLEGAPGETPEERINFO

Get Gap information of the peer device	
AT+BLEGAPGETPEERINFO= <conidx>,<info>	<conidx>, connection index <info>, information index 02: name 03: device version 04: device features 05: device appearance 06: slave preferred parameters 07: connection RSSI 08: connection channel map 24: PHY configuration 25: channel selection algorithm

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2.46 AT+BLEGAPPARNEGO

Negotiate the connection parameter with the peer device	
AT+BLEGAPPARNEGO= <conidx>, <intv_min>, <intv_max>, <latency>, <sup_to>	<conidx>, connection index (6~3200) <intv_min>, minimum interval (3~3200) <intv_max>, maximum interval (3~3200) <latency>, slave latency (0~499) <sup_to> supervision timeout (10~3200)

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3 AT Event List

3.1 BLEGAPCONN

Forward this event to the MCU after the Bluetooth LE connection is established.	
+BLEGAPCONN: <conn_idx>	<conn_idx>, connection index after connection established.

3.2 BLEGAPDISCONN

Forward this event to the MCU after the Bluetooth LE connection is disconnected.	
+BLEGAPDISCONN: <conn_idx>, <reason>	<conn_idx>, connection index. <reason>, reason code after disconnected, value is in hex.

3.3 BLEGATTFROMBLE

+BLEGATTFROMBLE: <conn_idx>, <att_idx>, <action>, <data>	<conn_idx>, the <conn_idx> returns from the *BLEGAPCONN* event after the connection is established. <att_idx>, it uses the static <att_idx> if the default ATT server is available. 01: Characteristic 1 03: Characteristic 2 <action> 00: Read 01: Write 02: Notification 03: Indication 04: Read Confirm 05: ATT info 06: ATT info Confirm 07: Notification (Empty) 08: Indication (Empty)

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Result:

```
AT+BLEGATTADDSERVICE=0000,41746D6F736963000000000000000F1
+BLEGATTADDSERVICE:0
OK

AT+BLEGATTADDCHAR=1800,41746D6F736963000000000000000F2,512
+BLEGATTADDCHAR:2
OK

AT+BLEGATTADDDESC=2902
+BLEGATTADDDESC:3
OK

AT+BLEGAPINIT=0F
OK

AT+BLEADVENABLE=0,ON
OK

# use nRF Connect app to make connection
+BLEGAPCONN:0

# use nRF Connect app to write string to ATT index 2
+BLEGATTFROMBLE:0,2,01,41746D6F736963
```

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3.4 BLEGATTCOMPEVT

"+BLEGATTCOMPEVT: <conn_idx>, <action>, <status>, <seq_num>"	<conn_idx>, the <conn_idx> returns from the *BLEGAPCONN* event after the connection is established. <action> 00: Read 01: Write 02: Notification 03: Indication 04: Read Confirm 05: ATT info 06: ATT info Confirm 07: Notification (Empty) 08: Indication (Empty) <status>, Please refer to 4 Status Code <seq_num>

Result:

```
AT+BLEGATTADDSERVICE=0000,41746D6F736963000000000000000F1
+BLEGATTADDSERVICE:0
OK

AT+BLEGATTADDCHAR=1800,41746D6F736963000000000000000F2,512
+BLEGATTADDCHAR:2
OK

AT+BLEGATTADDDESC=2902
+BLEGATTADDDESC:3
OK

AT+BLEGAPINIT=0F
OK

AT+BLEADVENABLE=0,ON
OK

# use nRF Connect app to make connection
+BLEGAPCONN:0
```

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```
# use nRF Connect app to enable CCCD
+BLEGATTFROMBLE:0,3,01,0100

AT+BLEGATTTOBLE=0,2,02,000102030405060708090A0B0C0D0E0F
OK
+BLEGATTTOBLE:0,02,0,0
```

3.5 BLEGATTMTUEXCHANGE

"+BLEGATTMTUEXCHANGE: <conn_idx>, <mtu>"	<conn_idx>, the <conn_idx> returns from the *BLEGAPCONN* event after connection is established. <mtu>, ATT MTU size

Result:

```
AT+BLEGAPINIT=0F
OK

AT+BLEADVENABLE=0,ON
OK

# use nRF Connect app to make connection
+BLEGAPCONN:0

# Apply MTU size (147) on nRF Connect app
+BLEGATTMTUEXCHANGE:0,147
```

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3.6 BLEGAPADVRPT

+BLEGAPADVRPT: <type>, <info>, <rssi>, <tx_pwr>, <phy_1>, <phy_2>, <addr_type>, <addr>, <data payload>	<type>, ADV type 0x00: Ext ADV 0x01: Legacy ADV 0x02: Ext Scan response 0x03: Legacy Scan response 0x04: Periodic ADV <info>, ADV information 0x1: Report is complete 0x2: Connectable. 0x4: Scannable. 0x8: Directed. <rssi>, rssi value between -127 and +20 <tx_pwr>, tx power in dBm <phy_1>, primary PHY (1: 1M, 2: 2M, 3: Coded) <phy_2>, secondary PHY (1: 1M, 2: 2M, 3: Coded) <addr_type>, address type (0: public, 1: private) <addr>, BD address of the advertising device <data payload> advertising report

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3.7 BLEGATTNOTI

Received GATT notification from peer device	
+BLEGATTNOTI: <data_array>	<data_array> received data

3.8 BLEGATTINDI

Received GATT indication from peer device	
+BLEGATTINDII: <data_array>	<data_array> received data

3.9 BLESMPPAIRENDIND

Received GATT indication from peer device	
+BLESMPPAIRENDIND: <conidx>, <status>	<conidx>, connection index <status> pairing end status

3.10 BLEGAPPAR

Received GAP parameter updated event	
+BLESMPPAIRENDIND: <conidx>, <intv>, <latency>, <sup_to>	<conidx>, connection index <intv>, connection interval <latency>, slave latency <sup_to> supervision timeout

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4 Status Code

There are two kinds of status code in the Atmosic AT command, AT Error Code and Application Error Code. The Atmosic AT parser will check the input content and will return the AT Error Code if the input contains wrong arguments. Figure 4-1 is the example if the user input the wrong <role> for the *AT+BLEGAPINIT* command.

```
AT+BLEGAPINIT=?
+BLEGAPINIT:<role>
OK

AT+BLEGAPINIT=00
ERR:6
```

Figure 4-1 AT Error Code Example

It will return an Application Error Code if an error occurs after applying the AT command. Figure 4-2 shows an example if the user did not apply the AT+*BLEGAPINIT* command before applying the AT+*BLEADVENABLE* command.

```
AT+BLEADVENABLE=?
+BLEADVENABLE:<act_idx>,<ON|OFF>
OK

AT+BLEADVENABLE=0,ON
+BLEADVENABLE:72
ERR
```

Figure 4-2 Application Error Code Example

4.1 AT Error Code

```
AT_ERR_NO_ERROR (0)
AT_ERR_CMD_NOT_SUPPORT (1)
AT_ERR_WRONG_ARGU_CNT (2)
AT_ERR_WRONG_ARGU_RANGE (3)
AT_ERR_WRONG_ARGU_TYPE (4)
AT_ERR_WRONG_ARGU_TYPE_OR_RANGE (5)
AT_ERR_WRONG_ARGU_CONTENT (6)
AT_ERR_CMD_SPECIFIC_ERR (128)
```

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4.2 Application Error Code

4.2.1 Atmosic Error Code

The following error code is also described in relating AT commands.

```
AT_CMD_NO_ERROR (0)
AT_GAP_ROLE_INITIALIZED (1)
AT_GAP_ROLE_NOT_SUPPORT (2)
AT_GAP_NEED_SET_BEFORE_GAP_INIT (3)
AT_SYS_NEED_REBOOT (4)
```

4.2.2 ATT Error Code

```
ATT_ERR_NO_ERROR (0)
ATT_ERR_INVALID_HANDLE (1)
ATT_ERR_READ_NOT_PERMITTED (2)
ATT_ERR_WRITE_NOT_PERMITTED (3)
ATT_ERR_INVALID_PDU (4)
ATT_ERR_INSUFF_AUTHEN (5)
ATT_ERR_REQUEST_NOT_SUPPORTED (6)
ATT_ERR_INVALID_OFFSET (7)
ATT_ERR_INSUFF_AUTHOR (8)
ATT_ERR_PREPARE_QUEUE_FULL (9)
ATT_ERR_ATTRIBUTE_NOT_FOUND (10)
ATT_ERR_ATTRIBUTE_NOT_LONG (11)
ATT_ERR_INSUFF_ENC_KEY_SIZE (12)
ATT_ERR_INVALID_ATTRIBUTE_VAL_LEN (13)
ATT_ERR_UNLIKELY_ERR  (14)
ATT_ERR_INSUFF_ENC (15)
ATT_ERR_UNSUPP_GRP_TYPE (16)
ATT_ERR_INSUFF_RESOURCE (17)
ATT_ERR_APP_ERROR (128)
```

4.2.3 GAP Error Code

```
GAP_ERR_NO_ERROR (0)
```

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```
GAP_ERR_INVALID_PARAM (64)
GAP_ERR_PROTOCOL_PROBLEM (65)
GAP_ERR_NOT_SUPPORTED (66)
GAP_ERR_COMMAND_DISALLOWED (67)
GAP_ERR_CANCELED (68)
GAP_ERR_TIMEOUT (69)
GAP_ERR_DISCONNECTED (70)
GAP_ERR_NOT_FOUND (71)
GAP_ERR_REJECTED (72)
GAP_ERR_PRIVACY_CFG_PB (73)
GAP_ERR_ADV_DATA_INVALID (74)
GAP_ERR_INSUFF_RESOURCES (75)
GAP_ERR_UNEXPECTED (76)
GAP_ERR_MISMATCH (77)
```

4.2.4 GATT Error Code

```
GATT_ERR_INVALID_ATT_LEN (80)
GATT_ERR_INVALID_TYPE_IN_SVC_SEARCH (81)
GATT_ERR_WRITE (82)
GATT_ERR_SIGNED_WRITE (83)
GATT_ERR_ATTRIBUTE_CLIENT_MISSING (84)
GATT_ERR_ATTRIBUTE_SERVER_MISSING (85)
GATT_ERR_INVALID_PERM (86)
```

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5 Examples

5.1 Use Default ATT Server

```
# Query current default GATT server state
AT+BLEGATTDFTSERVER?
+BLEGATTDFTSERVER:OFF
OK

# Turn ON default GATT server
AT+BLEGATTDFTSERVER=ON
+BLEGATTDFTSERVER:4
OK

# Reset system
AT+SYSRESET=00

# Query current default GATT server state
AT+BLEGATTDFTSERVER?
+BLEGATTDFTSERVER:ON
OK

# GAP initial
AT+BLEGAPINIT=0F
OK

# Enable advertisement
AT+BLEADVENABLE=00,ON
OK

# use nRF Connect app to make connection
+BLEGAPCONN:0
```

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5.2 Set Default ATT Server

```
# Query current default GATT server state
AT+BLEGATTDFTSERVER?
+BLEGATTDFTSERVER:ON
OK

# Set default service
AT+BLEGATTDFTSERVICE=0000,41746D6F7369630000000000000ABC1
OK

# Set default characteristic 1
AT+BLEGATTDFTCHAR1=1800,41746D6F7369630000000000000ABC2
OK

# Set default characteristic 2
AT+BLEGATTDFTCHAR2=1800,41746D6F7369630000000000000ABC3
OK

# Reset system
AT+SYSRESET=00
```

5.3 Turn Off Default ATT Server

```
# Query current default GATT server state
AT+BLEGATTDFTSERVER?
+BLEGATTDFTSERVER:ON
OK

# Turn OFF default GATT server
AT+BLEGATTDFTSERVER=OFF
+BLEGATTDFTSERVER:4
OK

# Reset system
AT+SYSRESET=00
```

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5.4 Set Advertisement

```
# Set advertising interval and duration
AT+BLEADVLEGACYPARM=0,32,64,0
OK

# Set advertising data
AT+BLEADVDATA=0,050841424344
OK

# Set scan response data
AT+BLESCANRSPDATA=0,03084154
OK

# GAP initial
AT+BLEGAPINIT=0F
OK

# Enable advertisement
AT+BLEADVENABLE=0,ON
OK
```

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5.5 Dynamic Add Service (Before BLEGAPINIT)

```
# Query current default GATT server state
AT+BLEGATTDFTSERVER?
+BLEGATTDFTSERVER:ON
OK

# Add service
AT+BLEGATTADDSERVICE=0000,41746D6F736963000000000000000001
+BLEGATTADDSERVICE:4
OK

# Add characteristic
AT+BLEGATTADDCHAR=1200,41746D6F736963000000000000000002,512
+BLEGATTADDCHAR:6
OK

# Add descriptor
AT+BLEGATTADDDESC=2902
+BLEGATTADDDESC:7
OK

# GAP initial
AT+BLEGAPINIT=0F
OK

# Enable advertisement
AT+BLEADVENABLE=0,ON
OK

# use nRF Connect app to make connection
+BLEGAPCONN:0
```

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5.6 Dynamic Add Server (After BLEGAPINIT)

```
# Query current default GATT server state
AT+BLEGATTDFTSERVER?
+BLEGATTDFTSERVER:ON
OK

# GAP initial
AT+BLEGAPINIT=0F
OK

# Enable advertisement
AT+BLEADVENABLE=0,ON
OK

# use nRF Connect app to make connection
+BLEGAPCONN:0

# Add service
AT+BLEGATTADDSERVICE=0000,41746D6F736963000000000000000001
+BLEGATTADDSERVICE:4
OK

# Add characteristic
AT+BLEGATTADDCHAR=1200,41746D6F736963000000000000000002,512
+BLEGATTADDCHAR:6
OK

# Add descriptor
AT+BLEGATTADDDESC=2902
+BLEGATTADDDESC:7
OK

# Active service and will see the service after refresh service in Nordic app
AT+BLEGATTSVCACTIVE=00
OK
```

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5.7 Notification

```
# Query current default GATT server state
AT+BLEGATTDFTSERVER?
+BLEGATTDFTSERVER:ON
OK

# Add service
AT+BLEGATTADDSERVICE=0000,41746D6F736963000000000000000001
+BLEGATTADDSERVICE:4
OK

# Add characteristic
AT+BLEGATTADDCHAR=1200,41746D6F736963000000000000000002,512
+BLEGATTADDCHAR:6
OK

# Add descriptor
AT+BLEGATTADDDESC=2902
+BLEGATTADDDESC:7
OK

# GAP initial
AT+BLEGAPINIT=0F
OK

# Enable advertisement
AT+BLEADVENABLE=0,ON
OK

# Use nRF Connect app to make connection
+BLEGAPCONN:0

# Read ccc descriptor from nRF Connect app and wait for response
+BLEGATTFROMBLE:0,7,00,

# Apply read confirm
AT+BLEGATTTOBLE=0,7,04,0100
OK
```

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```
# Send notification to nRF Connect app through att index (2)
AT+BLEGATTTOBLE=00,06,02,000102030405060708090A0B0C0D0E0F
OK

# Notification sent even
+BLEGATTTOBLE:0,02,0,0
```

6 How To Add New AT Command

There is an example AT command - "*custdemo*" in the BLE_atcmd example to demonstrate how to add a new AT command, and the source code located in "*BLE_atcmd/src/non_bt/custdemo.c*". This section will use the "custdemo" to describe how to add the new AT command.

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6.1 Source Code Format

The AT command consists of 6 elements *CMD_NAME*, *CMD_PARM_FMT*, *CMD_PARM_DESC*, *CMD_PARM_NUM*, *fn_cm_handler* and *AT_COMMAND*. See Table 6-1 and Figure 6-1.

AT Command Element	Description
CMD_NAME	AT command name
CMD_PARM_FMT	AT command format, please refer to AT command format (CMD_PARM_FMT section.
CMD_PARM_DESC	AT command description
CMD_PARM_NUM	Number of parameters of CMD_PARM_FMA
fn_cmd_handler	The AT command function callback after the user applies the correct CMD_NAME and CMD_PARM_FMT, and the developer needs to implement the feature in this function.
AT_COMMAND	Mandatory parameter, apply the AT command to system

Table 6-1 - AT Command Elements

```c
/**
 ****************************************************************************************
 *
 * @file custdemo.c
 *
 * @brief AT Command - Customer Demo Command
 *
 * Copyright (C) Atmosic 2021
 *
 ****************************************************************************************
 */

#include "at_cmd.h"
#include "atm_log.h"

#define CMD_NAME "CUSTDEMO"
#define CMD_PARM_FMT "B(0~1)"
#define CMD_PARM_DESC "<action>"
#define CMD_PARM_NUM 1

ATM_LOG_LOCAL_SETTING(CMD_NAME, V);

/**
 * AT COMMAND HANDLER
 ****************************************************************************************
 */

static void fn_cmd_handler(at_cmd_param_t *param)
{
    ATM_LOG(V, "%s: at_type (%d), err (%d)", __func__, param->type, param->err);

    if (param->err != AT_CMD_ERR_NO_ERROR) {
        return;
    }

    if (param->type == at_cmd_type_exec) {
        switch (param->args[0]->u8) {
            case 0: {
                ATM_LOG(V, "%s: AT command demo %u", __func__,
                    param->args[0]->u8);
            } break;

            case 1: {
                ATM_LOG(V, "%s: AT command demo %u", __func__,
                    param->args[0]->u8);
            } break;

            default: {
                ATM_LOG(V, "%s: Not support %u", __func__, param->args[0]->u8);
            } break;
        }
    }
}

AT_COMMAND(CMD_NAME, CMD_PARM_FMT, CMD_PARM_NUM, fn_cmd_handler, CMD_PARM_DESC);
```

Figure 6-1 - AT Command Elements

6.2 AT Command Format (CMD_PARM_FMT)

Notation	Meaning	Source	Description
b(#~@)	int8_t	ASCII	
B(#~@)	uint8_t	ASCII	B(1~2), unsigned 8 bit number from 1 to 2.
w(#~@)	int16_t	ASCII	
W(#~@)	uint16_t	ASCII	W(~5), unsigned 16 bit number from 0 to 5.
d(#~@)	int32_t	ASCII	d(-3~10), signed 32 bit number from -3 to 10 d(~1000), signed 32 bit number from -2147483647 to 1000
D(#~@)	uint32_t	ASCII	D(30~), unsigned 32 bit number from 30 to 4294967295
A(#~@)	byte array	ASCII	A, byte array with any length A(3~), byte array with at least 3 bytes A(2~5), byte array with length from 2 to 5 octets A(6~6), byte array with 6 bytes exactly
S(#~@)	UTF8 string	UTF8	S, string with length 0 to 255 octets S(3~), string with length from 3 to 255 octets S(2~5), string with length from 2 to 5 octets

#: minimal number
@: maximal number

Table 6.2-1 AT Command Format

6.3 Modify Makefile

Add the source file into makefile after creating the AT command and rebuild the firmware by using “make clean” and “make run_all”. See Figure 6.3-1.

```
############################ Example(application) Sources ######################
SRC_TOP = src
SRC_NONBT = src/non_bt
SRC_BT = src/bt

CFLAGS += -include $(SRC_TOP)/atcmd_version.h

# application : Non-BT part
INCLUDES += $(SRC_NONBT)

# application: BT part
INCLUDES += $(SRC_BT)
C_SRCS += $(SRC_BT)/atcmd_gap.c

# application: non-BT part
C_SRCS += $(SRC_NONBT)/custdemo.c

INCLUDES += $(SRC_TOP)

UU_TEST := BLE_atcmd atcmd_gap custdemo

# Default NVDS
flash_nvds.data := \
    11-SLEEP_ENABLE/hib \
    12-EXT_WAKEUP_ENABLE/enable \

###############################################################################
include $(COMMON_USER_DIR)/profiles.mk
include $(COMMON_USER_DIR)/framework.mk
```

Figure 6.3 -1 Modify Makefile

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7 Atmosic Remote Control Functional Tester

7.1 How to build the tester

The BLE_atcmd application can also be used to build the firmware of the Atmosic Remote Control Functional Tester with following command:

make PCBA=pcb057 USER_BD_ADDR="xx xx xx xx xx xx" run_all

Basically, the functional tester cooperates with the Production test PC tool by AT commands through UART0. But the user still could use the terminal SW to perform some handy tests.

7.2 UART0 interface configuration

Note that the UART0 baud rate is 460800 with HW flow control enabled. 8 bits, no parity and 1 stop bits are the rest settings that you might need to notice.

7.3 Example usage

Users may reference the following table to start an ATVV audio stream on the Atmosic Remote Reference design and capture the audio packets:

Operations	AT command	Description
Initialization	=>AT+BLEGAPINIT=05 <=OK	05: client role
SCAN Filter Add	=>AT+BLESCANFILADD=0,9,41746d6F736963205 24355 <=OK	09: name tag 41…: Atmosic RCU

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Operations	AT command	Description
SCAN ON/OFF	=>AT+BLESCANENABLE=0,-50,ON <=OK =>AT+BLESCANENABLE=0,-50,OFF <=OK	0: scan idx -50: min rssi
ADV Report	+BLEGAPADVRPT:1,7,-60,127,1,0,0,010812231118,020106030312180319800104ff0a24020c0941746d6f73696320524355	0: address type 010812231118: address(lsb first)
Create connect	AT+BLEGAPCREATECONN=0,0,181123120801 OK +BLEGAPCONN:0	0: initiator index 0: address type 181123120801: address(msb first) 0:status

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Operations	AT command	Description
PAIRING	AT+BLESMPPAIRREQ=0,0,0,0,16,7,7 OK +BLESMPPAIRENDIND:0,0	0: connect index 0,0,0: no input no output/no oob/no bonding 16: key size 16 7,7: all key dist. 0: status
Enable HID report notification by writing CCCD	AT+BLEGATTWR=0,44,0100 OK +BLEGATTWR:0,0	0: connect index 44: handle of cccd (this was set by UI) 0100: Value of notifying enabled 0: status
Received Key	+BLEGATTNOTI:0,42,ea000000 +BLEGATTNOTI:0,42,00000000	0: connect index 42: handle of report ea000000: report value

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Operations	AT command	Description
ATVV service finding for handle range	=>AT+BLEGATTDISCSVC=0,ab5e00015a214f05bc7daf01f617b664 <=OK <=+BLEGATTDISCSVC:0,0,49,57,64b617f601af7dbc054f215a01005eab	49,57: handle range of ATVV service.
ATVV chars parsing for handles	=>AT+BLEGATTDISCCHARS=0,49,57,50 <=OK =>+BLEGATTDISCCHARS:0,10,3,8,50,51,64b617f601af7dbc054f215a02005eab,18,52,53,64b617f601af7dbc054f215a03005eab,18,55,56,64b617f601af7dbc054f215a04005eab	51:TX Handle 53:RX Handle 56:CTL Handle
ATVV descs parsing for CCCD	=>AT+BLEGATTDISCDESCS =0,49,57,50 <=OK <=+BLEGATTDISCDESCS:0,10,9,49,0028,50,0328,51,64b617f601af7dbc054f215a02005eab,52,0328,53,64b617f601af7dbc054f215a03005eab,54,0229,55,0328,56,64b617f601af7dbc054f215a04005eab,57,0229	54: RX cccd handle 57:CTL cccd handle

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Operations	AT command	Description
Enable ATVV notifications by writing CCCD	=>AT+BLEGATTWR=0,54,0100 <=OK =>+BLEGATTWR:0,0 =>AT+BLEGATTWR=0,57,0100 <=OK =>+BLEGATTWR:0,0	0: connect index 54: RX cccd handle 57: CTL cccd handle 0100: Value of notifying enabled 0: status
ATVV 0.4 get capability	=>AT+BLEGATTWR=0,51,0a00040003 <=OK <=+BLEGATTWR:0,0 <=+BLEGATTNOTI:0,56,0b0004000300860014 note: If DUT is ATVV 1.0, we would recevie “+BLEGATTWR:0,3”.	51: Tx handle 0a: Get cap 0004: 0.4 0003: 8K/16K 56: Ctl handle 0b: Get cap response 0086: audio frame size
ATVV 1.0 get capability	=>AT_BLEGATTWRWO=0,51,0a0100000300 <=OK <=+BLEGATTWRWO:0,0 <=+BLEGATTNOTI:0,56,0b0100030000800100	51: Tx handle 0a: Get cap 0100: 1.0 0003: 8K/16K 00: on-request 56: Ctl handle

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Operations	AT command	Description
		0b: Get cap response 0080: audio frame size 01: enable DLE 00: not used
ATVV 0.4 mic open	=>AT+BLEGATTWR=0,51,0C0003 <=OK <=+BLEGATTWR:0,0 <=+BLEGATTNOTI:0,56,04 <=+BLEGATTNOTI:0,53,000001000213ffff7fffffa7b0 8808808807b87c08808807c08808808807b72f6f080 80080080000800000818100000010011011111111 2 12123232342334233423342334233432342422323342 33431422334234234224 1bf7f7777700800f3f5080008 00000000080008000008f3000ff50f58f7808808808 8088808888880890988888988998	51: Tx handle 0C: Mic open 0003: 8K/16K 56: Ctl handle 04: Audio start 53: Rx handle …: audio data
ATVV 1.0 mic open	=>AT+BLEGATTWRWO=0,51,0C00 <=OK <=+BLEGATTWRWO:0,0 <=+BLEGATTNOTI:0,56,04000200 <=+BLEGATTNOTI:0,53,ffff7fffffa7b08808808807b8 7c08808807c08808808807b72f6f080800800008000 8000000081000000101010111021112213222333342 33343334233423334234233242334233242333352 3 233342423323433342334225202313351314132142	51: Tx handle 0C: Mic open 00:Playback 04: Audio start 00: reason(mic open) 02: 16K/16bits

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Operations	AT command	Description
	22306230023235102420383683839583840959 32a419311a520a333b35009	00: audio stream id
ATVV 0.4 mic close	=>AT+BLEGATTWR=0,51,0D <=OK <=+BLEGATTWR:0,0 <=+BLEGATTNOTI:0,56,00	51: Tx handle 0D: Mic close 56: Ctl handle 00: Audio end
ATVV 1.0 mic close	=>AT+BLEGATTWRWO=0,51,0D00 <=OK <=+BLEGATTWRWO:0,0 <=+BLEGATTNOTI:0,56,0000	51: Tx handle 0D: Mic close 00: Stream id 56: Ctl handle 00: Audio end 00: reason(mic_close)

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